# Pascal's Triangle (1641)

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

- **Corollary**

Pascal's identity, together with the initial conditions $\binom{n}{0} = \binom{n}{n} = 1$ for all integers *n*, can be used to give a geometric arrangement of the binomial coefficients in a triangle, called Pascal's triangle, as follows:

$$\binom{0}{0}$$

$$\binom{1}{0}\ \binom{1}{1}$$

$$\binom{2}{0}\quad\binom{2}{2}$$

$$\binom{3}{0}\qquad\binom{3}{3}$$

$$\binom{4}{0}\qquad\binom{4}{4}$$

$$\binom{5}{0}\qquad\binom{5}{5}$$

$$\binom{6}{0}\qquad\binom{6}{6}$$

$$\binom{7}{0}\qquad\binom{7}{7}$$

$$\binom{8}{0}\qquad\binom{8}{8}$$

